

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460870.8692 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.058
Transit depth (Rp/Rs)^2	3.13 +/- 2.06 [%]
Orbital Inclination (inc)	84.77 +/- 1.61
Airmass coefficient 1 (a1)	1.54 +/- 0.12
Airmass coefficient 2 (a2)	-0.19 +/- 0.69
Scatter in the residuals of the lightcurve fit is	7.58 %
Best Comparison Star	#1 - [595, 288]
Optimal Aperture	1.83
Optimal Annulus	13.9
Transit Duration (day)	0.036 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.058 vs 0.14762 (deviation 0.51σ)
Duration fit vs published	0.036 d vs 0.1033 d (deviation 1.92σ)
Mid-time O–C	-3.2 min
Depth SNR	3.1
Residual scatter	7.58 %

Light curve

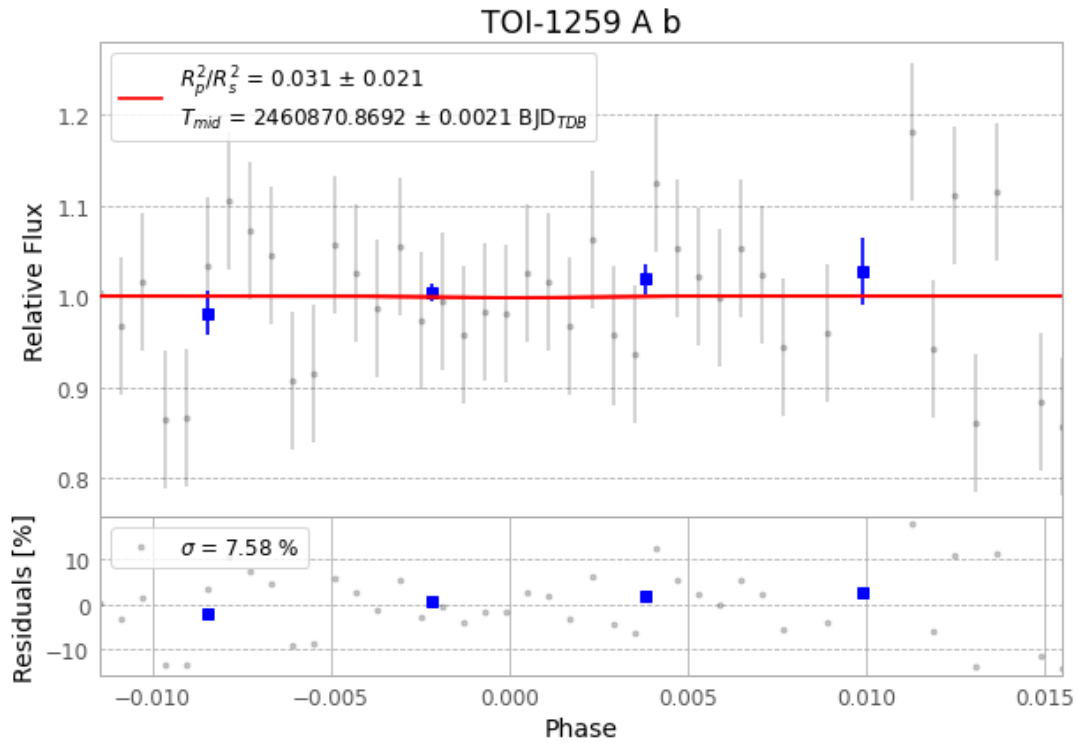


Figure 1 — FinalLightCurve_TOI-1259 A b_2025-07-14.png (SHA-256 7945a1d9527d5f5b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d1b4f431cba40417...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), TOI1259250714075417.FITS → TOI1259250714111216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-250714.log (8b54bba39bd8...), FinalLightCurve_TOI-1259 A b_2025-07-14.png (7945a1d9527d...), FinalLightCurve_TOI-1259 A b_2025-07-14.csv (93fb14bb561a...)

Compute resources

Wall time 324.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:29Z.